Arista Access Node IND-ACC-16 Indiranagar Exchange QoS Congestion Event Resolution SOP

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7110

Description

This Standard Operating Procedure (SOP) addresses the failure domain of QoS congestion events, specifically microbursts and interface tail drops, on the Arista Access Node IND-ACC-16 at the Indiranagar Exchange. The root cause of this issue lies within the interplay between the node's buffer management and the configured Quality of Service (QoS) policies. The Arista Access Node employs an architecture that utilizes a shared buffer space for all ingress and egress traffic. However, when QoS policies prioritize certain traffic flows over others, it may lead to a scenario where the buffer space becomes saturated, resulting in microbursts and interface tail drops. Furthermore, the QoS queue starvation occurs when a particular queue is not allocated sufficient bandwidth, causing packets to be dropped. This SOP aims to identify and resolve the root cause of the issue by analyzing the QoS configuration, buffer management settings, and monitoring the network for signs of congestion.

Pre-requisite Checks

- Physical Checks: Dispatch remote hands to the Indiranagar Exchange to verify the node's power and connectivity status.

- Environmental Checks: Check the ambient temperature and humidity levels around the node to ensure they are within the recommended operating range.

- Network Checks: Verify the redundant link status and ensure it is operational to prevent any potential data loss during the troubleshooting process.

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Buffer management settings	Medium
Hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface statistics, Expected Output: Interface statistics, including packet drop counts and buffer utilization., Description: This command helps identify the interface experiencing the highest packet drop counts, indicating potential congestion issues.

Command: show qos configuration, Expected Output: Current QoS configuration, including policy maps and class maps., Description: This command provides insight into the configured QoS policies and
```

```
helps identify potential mismatches or conflicts.

Command: show buffers, Expected Output: Buffer utilization statistics, including shared and per-interface buffers., Description: This command helps analyze the buffer space allocation and utilization, providing clues about potential buffer management issues.
```

Resolution Steps

Step 1: Step: Verify QoS configuration, Command: show qos configuration, Description: Analyze the current QoS configuration to ensure it is correctly implemented and does not cause any mismatches or conflicts.

Step 2: Step: Adjust buffer management settings, Command: buffers shared 1024, Description: Adjust the shared buffer space allocation to prevent buffer saturation and reduce the likelihood of microbursts and interface tail drops.

Step 3: Step: Re-configure QoS policies, Command: qos policy-map , Description: Re-configure the QoS policies to ensure proper prioritization and allocation of bandwidth to prevent queue starvation.

Post-Check Verification

```
Command: show interface statistics, Expected Output: Improved interface statistics, with reduced packet drop counts and buffer utilization., Description: Verify that the diagnostic commands show an improvement in interface statistics, indicating the resolution steps have effectively addressed the congestion issue.

Command: show qos configuration, Expected Output: Updated QoS configuration, with corrected policy maps and class maps., Description: Verify that the QoS configuration has been updated to prevent any potential mismatches or conflicts.

Command: show buffers, Expected Output: Improved buffer utilization statistics, with reduced buffer saturation., Description: Verify that the buffer management settings have been adjusted to prevent buffer saturation and reduce the likelihood of microbursts and interface tail drops.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: qos policy-map , Description: Restore the original QoS configuration to prevent any potential mismatches or conflicts.

- Step: Revert buffer management settings, Command: buffers shared , Description: Revert the buffer management settings to their original values to prevent buffer saturation.

- Step: Disconnect redundant link, Command: interface gigabitEthernet 0/0/0, Description: Disconnect the redundant link to prevent any potential data loss during the rollback process.